Quantum Physics 1

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Abstract

This course was taught by Dr. Liam Fitzpatrick at Boston University during the Spring 2025 semester. We used the text Quantum Mechanics: A Paradigms Approach by David McIntyre. We also use the course website physics.bu.edu/~fitzpatr.

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Chapter 1

Introduction

Lecture 1: Stern-Gerlach Experiments

Tue 21 Jan 2025 14:00

The Stern-Gerlach experiments were done in the 20s, while quantum physics was still being developed. We are given a set of silver atoms, which we will consider for this experiment to be an extremely tiny bar magnet. It has a dipole magnetic moment (vector) $\boldsymbol{\mu}$. The experiment takes a large magnet (image in notes, ill draw later) with a nonconstant magnetic field between north and south. The silver atom is then shot through the $\mathbf{B}$ field, and we look at where it comes out.

Classical mechanics says the magnetic moment $\boldsymbol{\mu}$ passing through the field $\mathbf{B}$ should deflect it as follows. Energy is given by

$$E = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

We then get the force

$$\mathbf{F} = -\nabla E.$$

For example, if $\mathbf{B} = B_z(z)\hat{\mathbf{z}}$ is slightly varying along the z direction, then

$$\mathbf{F} = -\partial_z B_z \mu_z.$$

Note here that ∂_z is the partial wrt z. New notation!!!

The force then depends only on the component of the magnetic moment in the z-direction. $F \propto \boldsymbol{\mu} \cdot \hat{\mathbf{z}} = |\boldsymbol{\mu}| \cos\theta$, where θ is the angle formed with the z-axis. So

$$F_z = |\boldsymbol{\mu}| \frac{\partial B_z}{\partial z} \cos\theta.$$

Since the angle of the moments is distributed randomly, we would expect the distribution of resultant locations to be constant, since each magnetic moment corresponds to a new location, and since the angles should be distributed evenly in θ, they should be distributed evenly through the locations. However, even if they weren't *purely* random, we would get *some distribution.* It would be weird if we didn't see a distribution. So of course we don't see a distribution, because physics.

When we actually do the experiment, they only ever come out at two points, a top point and a bottom point. Since we know what $B_z(z)$ is, and since we know the mass of a silver atom, we can find the force and thus find the exact magnetic moment. We find that $\mu_z = \pm |\boldsymbol{\mu}|$ always. Even though the angles were randomly distributed, it always shows up as two possibilities.

We say it is quantized. In classical mechanics, we have a continuous distribution, but in quantum mechanics, the possibilities are all quantized into two possibilities. In quantum physics, we will talk a lot about the state of a system, and $|\Psi\rangle$ will denote the state of some system. The symbol $|\cdot\rangle$ is called a "ket".

Classically, the state is the direction of the magnetic moment, since that determines all the information we need. However, in quantum mechanics, we cannot describe the state of a system using classical physics, which is why we use kets to denote it. They are a completely new thing that doesn't show up in classical mechanics, and we will describe precisely what a ket is as we continue.

Since there are only two possibilities in the experiment, there must be only two states. We can find that those two states are

- $|+\rangle$ is the state where $\mu_z = +|\boldsymbol{\mu}|$;
- $|-\rangle$ is the state where $\mu_z = -|\boldsymbol{\mu}|$;

where again $|\boldsymbol{\mu}|$ is the moment of a silver atom.

Another thing to note is that the distribution is even throughout the possible states; 50% of the time we see $|+\rangle$, and the rest of the time we measure a state of $|-\rangle$.

Experiment two takes all the atoms that have already been deflected, and passes them through *another* magnetic field. (draw later) All the up ones keep going up, and all the down ones keep going down. Basically we didn't change anything so it stays the same. Makes sense. Ok technically we only re-measure $|+\rangle$ states.

Experiment 3 starts off the same way as experiment 1, and then proceeds *similarly* to experiment 2, but instead of measuring them in the $\hat{\mathbf{z}}$ axis again, we change the magnetic field to point in the $\hat{\mathbf{x}}$ direction. The z direction isn't special or anything, so we can do this. Classically, we would note that $\mu_z = |\boldsymbol{\mu}|$, so we would assume that $\mu_x = 0$, since the entire moment is used up in the z-component. BUT of course we get a weird answer. We once again see exactly two spots hit; exactly 50% of the time they end up in the $+x$-direction, and 50% of the time they go to the $-x$-direction. So the conclusion is the moment has multiple distinct components; orthogonal directions correspond to distinct components. We call these states $|+\rangle_x$ and $|-\rangle_x$, respectively.

Experiment 3 continues by measuring the atoms once again. So we measured them in z and found them to be distributed 50/50 in two spots. We took the $|+\rangle$ state and measured them in the x-direction and found another 50/50 distribution. However, we already measured them as $|+\rangle$ in the z-direction, so we expect them to still be in this state. This is like if we took a black sock, sorted it based on whether it was striped or not, then checked to see if it was still black. However, we *DON'T SEE THIS.* We see instead another 50/50 distribution. The particles are no longer still $|+\rangle$, but instead it's back to a completely random distribution.

So what does this mean? Measuring in z repeatedly keeps it in the same state, but when we measured it in the x component, it changed. Later, we will explain how we know this is actually what happens, and how we know this isn't the experimental machine just being shit. So measuring it in one component changed the other component. Additionally, if we only change the direction a little, rather than orthogonally, we still measure a new distribution, it's just not 50/50. So it is the act of measurement itself that changes the state of the particle. In classical physics, measuring something doesn't change it. Welcome to quantum physics, where the state $|\Psi\rangle$ cannot have a well-defined magnetic moment in two directions at the same time.

One last point, which we will discuss later. The magnetic moment $\boldsymbol{\mu}$ is related to the spin $\mathbf{S}$ by the relation

$$\boldsymbol{\mu} = \frac{ge}{2m_e}\mathbf{S}.$$

The symbol g is some constant. The spin $\mathbf{S}$ is the angular momentum of the electron when it's sitting still. Moment pointing up correspnds to

$$S_z = \frac{\hbar}{2}.$$

Definition 1.1: Kronecker Delta

The Kronecker Delta is denoted δ_{ij}, and is defined as

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}.$$

We can interpret δ_{ij} as the identity matrix $\mathbb{1}$, since it has 1s only when the indices $i = j$. Let $\mathbf{a}$ be a vector. Then $\delta_{ij} a_j = a_i$ (in Einstein notation).

Lecture 2: Linear Algebra

Thu 23 Jan 2025 14:00

The state of a system encodes all information about a system. We denote this by $|\psi\rangle$ in quantum physics. Also note that when an index is shared over two elements:

$$M_{ik} N_{kr},$$

this implies this expression really indicates a summation over the shared index:

$$M_{ik} N_{kr} \coloneqq \sum_k M_{ik} N_{kr}.$$

A ket $|\psi\rangle$ represents the state of some system, but it is also a vector in a $\mathbb{C}$-vector space. This ket notation is useful since we may deal with infinite-dimensional spaces, and it also plays nicely with inner products. We always assume a vector space in quantum mechanics is a space in $\mathbb{C}$-**Vect**.

The question is, what really is $|\psi\rangle$, and how do we get information out of it? In order to do this, we need to equip our space with an inner product, making it a Hilbert space. Note that all n-dimensional $\mathbb{C}$-Hilbert spaces can be given as the matrix space $M_{n\times 1}(\mathbb{C})$. We then define $|\psi\rangle^\dagger \coloneqq \pi^{-1}(\pi(|\psi\rangle)^\dagger)$, for $\pi : \mathcal{H} \to M_{n\times 1}(\mathbb{C})$ the isomorphism. More simply, given a vector

$$\begin{pmatrix} x \\ y \end{pmatrix},$$

define

$$\begin{pmatrix} x \\ y \end{pmatrix}^\dagger = \begin{pmatrix} x^* & y^* \end{pmatrix}.$$

Equivalently,

$$(z_1|v_1\rangle + z_2|v_2\rangle)^\dagger = z_1^*\langle v_1| + z_2^*\langle v_2|.$$

The Hermitian conjugate $|\psi\rangle^\dagger$ is denoted as a 'bra', $\langle\psi|$. Note that given vectors $|\phi\rangle, |\psi\rangle$, we can define an inner product

$$|\phi\rangle \cdot |\psi\rangle \coloneqq \langle \phi \,|\, \psi\rangle.$$

In a more general vector space, we still denote the inner product as $\langle \phi \,|\, \psi\rangle$.

Definition 1.2: Inner Product

An inner product is a map $\langle \cdot \,|\, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ such that

- $\langle v_1 \,|\, v_2\rangle = \langle v_2 \,|\, v_1\rangle^*$;
- $\langle v \,|\, v\rangle \geq 0$;
- $\langle v \,|\, v\rangle = 0$ iff $|v\rangle = 0$;
- $(a\langle v_1| + b\langle v_2|) \cdot (c|w_1\rangle + d|w_2\rangle) = ac\langle v_1 \,|\, w_1\rangle + ad\langle v_1 \,|\, w_2\rangle + bc\langle v_2 \,|\, w_1\rangle + bd\langle v_2 \,|\, w_2\rangle$.

Our way of pulling information out of a ket will be by taking its inner product with a state we already understand. For example, in the Stern-Gerlach experiments, we knew the state $|+\rangle$ indicated $\boldsymbol{\mu}$ pointing up, and $|-\rangle$ indicated pointing down. Assume $|+\rangle, |-\rangle, |\psi\rangle$ are all normalized with norm 1; $\langle \psi \mid \psi \rangle = 1$. For some reason, $|+\rangle, |-\rangle$ form a basis. We can then find the inner product of ψ with any vector using this fact. The Born Rule tells us what information this actually gives us; we will state it here, specialized to the Stern-Gerlach experiments.

Proposition 1.1

The probability of finding the state $|\psi\rangle = |+\rangle$ is

$$P_+ = |\langle + \mid \psi \rangle|^2 .$$

Similarly, the probability of finding the state $|\psi\rangle = |-\rangle$ is

$$P_- = |\langle - \mid \psi \rangle|^2 .$$

Moreover, when we measure $|\psi\rangle$ in one of these states, it turns into that state.

Since the probabilities have to sum to 1, we assume ψ has a unit norm. This follows since $\langle + \mid - \rangle = 0$, so the probability of finding it at both $|+\rangle$ and $|-\rangle$ is 0.

Since $|+\rangle, |-\rangle$ forms a complete basis, we can write

$$|\psi\rangle = \langle + \mid \psi \rangle |+\rangle + \langle - \mid \psi \rangle |-\rangle.$$

Then

$$\| |\psi\rangle \| = |\langle + \mid \psi \rangle|^2 + |\langle 0 \mid \psi \rangle|^2 .$$

Lecture 3: owo

Tue 28 Jan 2025 14:01

The kets $|+\rangle, |-\rangle$ form a complete basis because they are the only measurable states, and the vector space is the space of all states.

The born rule says

$$P_\pm = |\langle \pm \mid \psi \rangle|^2 .$$

That is, the probability of measuring a given state is the same as the norm squared of the inner product with the state vector ψ with the observable state $|\pm\rangle$.

We can also change the direction in which we measure from the $\hat{z}$ direction to the $\hat{x}$ direction. We write $|\pm\rangle_x$. We can assume WLOG that these states are normalized; that is, $\langle + \mid + \rangle_x = 1$. Moreover, $\langle + \mid - \rangle_x = 0$, because we cannot measure one state and then have it change in repeated experiments with no change. This means state vectors are orthogonal. Everything about the x-states is the same as the z-states.

It's difficult to describe how to desccibe a well-defined z and x direction at the same time; how can we show what direction $|\psi\rangle$ points in given only 2 states?

Note that if you measure $|+\rangle$ in the z-direction, then you have a 50% chance of measuring in $|+\rangle_x$ or $|-\rangle_x$, so we have that

$$P_+^{(x)} = P_-^{(x)} = \frac{1}{2}.$$

It follows that

$$|\psi\rangle_x = C_+^{(x)} |+\rangle_x + C_-^{(x)} |-\rangle.$$

In this case, we have

$$P_\pm^{(x)} = \frac{1}{2} = |\langle +_x \mid + \rangle|^2 = |\langle -_x \mid + \rangle|^2 ,$$

because measuring $|+\rangle$ gives a $1/2$ chance of measuring either $+_x$ or $-_x$. Since $C_{\pm}^{(x)} \in \mathbb{C}$, we have that

$$\left|C_{\pm}^{(x)}\right| = \frac{1}{\sqrt{2}}.$$

This is because

$$C_{\pm}^{(x)} = \langle \pm_x \,|\, + \rangle.$$

This follows from orthogonality. Since these coefficients are complex-valued, this only tells us the probability, but we cannot discern the phase of these states. The phase of a complex number z is the θ such that $z = re^{i\theta}$. We only know the magntidude, r. This is as far as we can go experimentally, and so we can arbitrarily choose values for $C_{\pm}^{(x)}$ without loss of generality, so we define

$$|+\rangle = \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x.$$

We can use the same logic to find that the magntitude of the coefficients

$$|-\rangle = C_{+}^{(x)}|+\rangle + C_{-}^{(x)}|-\rangle$$

are also $1/\sqrt{2}$. However, we also know that since $\langle + \,|\, - \rangle$ are orthogonal, we have that

$$\langle - \,|\, + \rangle = \frac{C_{+}^{(x)}}{\sqrt{2}}\langle +_x \,|\, +_x \rangle + \frac{C_{-}^{(x)}}{\sqrt{2}}\langle -_x \,|\, -_x \rangle = 0.$$

We choose the convention that

$$C_{+}^{(x)} = \frac{1}{\sqrt{2}}, \qquad C_{-}^{(x)} = -\frac{1}{\sqrt{2}}.$$

$$\boxed{\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x, \\ |-\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x - \frac{1}{\sqrt{2}}|-\rangle_x. \\ |+\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle, \\ |-\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle. \end{aligned}}$$

So if the act of measuring outputs some $|+\rangle$, since this doesn't have a definite x-component, it's then obvious how this x-state gets scrambled. This is superposition; a great way of saying this is "Schrodinger's cat is a state in a vector space which is a linear combination of dead and alive". This is because there is no way to describe what this physicsally means in english.

Let's review the third Stern-Gerlach experiment again. We have measurements

$$|\psi\rangle \longrightarrow Z \longrightarrow X \longrightarrow Z$$

We can give the isomorphic representations

$$|+\rangle \cong \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle \cong \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In this same representation, we have

$$|+\rangle_x \cong \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |-\rangle_x \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

We can also perform measurements in the Y direction, and use the same logic to determine the representations of $|\pm\rangle_y$. We have

$$|+\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle + i|-\rangle\right),$$
$$|-\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle - i|-\rangle\right).$$

Our representations are then

$$|+\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |-\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}.$$

Now we want to talk about the generalized Born's rule. Let $\{|a_i\rangle\}_{i\in I}$ be the collection of all possible obserable states. Since we cannot observe two different states, these vectors must be orthonormal:

$$\langle a_i \mid a_j\rangle = \delta_{ij}.$$

The probability

$$P_{a_i} = \left|\langle a_i \mid \psi\rangle\right|^2.$$

Lecture 4: Operators and Measurement

Tue 04 Feb 2025 14:00

Postulate: Every observation can be represented by an operator on the space of states. For example, the measurement operators in the Stern-Gerlach experiments are S_x, S_y, and S_z. The only things you can measure are the eigenvectors of the operator of an experiment.

$$S_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Clearly this has eigenvalues $\pm\frac{\hbar}{2}$ with eigenvalues as the standard basis for $\mathbb{R}^2$. Wow, that's $|+\rangle, |-\rangle$.

$$S_x = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

This has the same eigenvalues with eigenvectors

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix}, \qquad -\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}.$$

Wow, that's $|+\rangle_x, |-\rangle_x$! Also,

$$S_y = \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}.$$

Wow, this has the same eigenvalues but its eigenvectors are $|+\rangle_y, |-\rangle_y$. So we have thus noticed that the measurements we can observe are precisely the eigenvectors of the spin matrices.

Any hermitian operator A with eigenvalue/eigenvector pairs $a_i, |a_i\rangle$, then (assuming finite dimensionality) can be written as

$$A = \sum_{i=1}^{N} a_i |a_i\rangle\langle a_i|.$$

We can show this: node that eigenvectors with different eigenvalues are mutually orthogonal ($\langle a_i \mid a_j\rangle = \delta_{ij}$), so

$$A|a_i\rangle = \left(\sum_{j=1}^{N} a_j |a_j\rangle\langle a_j|\right)|a_i\rangle = a_i|a_i\rangle.$$

This is often the best representation for matrix operators, since we care mostly about their eigenvectors and eigenvalues. For example,

$$S_x = \frac{\hbar}{2}|+\rangle_x\langle+|_x - \frac{\hbar}{2}|-\rangle_x\langle-|_x = \frac{\hbar}{4}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} - \frac{\hbar}{4}\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

What if we consider a Stern-Gerlach machine oriented along the direction $\mathbf{n}$? We can normalize it and write it in sperical coordinates:

$$\hat{\mathbf{n}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

We are writing θ as the angle with z and ϕ as the angle in xy. Then we can write

$$\hat{\mathbf{n}} \cdot \mathbf{S} = n_x S_x + n_y S_y + n_z S_z.$$

Note that $\mathbf{S}$ is a vector of operators, so this is a largely symbolic representation.

Definition 1.3: Hermitian Operator

An operator M is *hermitian* if $M^\dagger = M$ as an operator.

In an abstract vector space, $\dagger$ is defined as the operator such that

$$\langle\phi|M^\dagger|\psi\rangle = (\langle\phi|M|\psi\rangle)^*.$$

Definition 1.4: Projection Operator

A projection operator is an operator P such that $P^2 = P$.

Examples include the operators

$$P_+ = \begin{pmatrix}1 & \\ & \end{pmatrix}, \qquad P_- = \begin{pmatrix} & \\ & 1\end{pmatrix}.$$

If we have any basis $\{|a_i\rangle\}$, then we can define

$$P_{a_i} := |a_i\rangle\langle a_i|.$$

Additionally,

$$\sum_i P_{a_i} = \mathbb{1}.$$

This is called the completeness relation.

Projection operators are how we define measurement. Suppose we perform a measurement and find the state $|a_i\rangle$. Then the final state after measurement is

$$\frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}}.$$

The new Born rule is then as follows. Let $\{a_i, s\}$ be the collection of all eigenvectors with eigenvalue a_i. Then

$$P_{a_i} = \sum_{s=1}^{N} |a_i, s\rangle\langle a_i, s|.$$